

# Mini Search Engine / Information-Retrieval System

**Data Structures and Algorithms – 3 (25CS2103E)**

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**Team 13      Section 13**

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## Abstract

The Mini Search Engine / Information-Retrieval System is designed to index a document corpus and efficiently answer four major query classes: exact keyword search, phrase/substring search, fuzzy “Did You Mean?” search, and document-similarity search.

- Inverted indexing enables fast keyword retrieval.
- Efficient string-matching techniques process phrase and substring queries.
- Edit distance identifies and ranks closely matching terms for fuzzy search.
- Term-frequency-based representations and similarity measures retrieve related documents.
- Core algorithms are implemented from scratch rather than using built-in search-engine libraries.

# Introduction

- The project demonstrates the core principles behind a small information-retrieval system.
- A document corpus is indexed so that user queries can be answered efficiently.
- The system combines indexing, string matching, fuzzy matching, ranking, and similarity techniques.
- The implementation emphasizes understanding the underlying Data Structures and Algorithms.

Source basis: project abstract

# Literature Survey

The supplied abstract does not list specific papers, authors, or prior systems. Therefore, this slide presents the project's identified technical areas rather than inventing citations.

- Information Retrieval – indexing and retrieval of documents.
- Inverted Index – efficient keyword-based retrieval.
- String Matching – phrase and substring search.
- Edit Distance – fuzzy matching and “Did You Mean?” suggestions.
- Term-Frequency Representation – representing documents for similarity comparison.
- Similarity Measures – retrieving documents related to a query or document.

# Problem Statement and Objectives

## Problem Statement

- Provide efficient search over a document corpus.
- Support different query types instead of only exact keyword matching.
- Demonstrate the underlying algorithms without relying on built-in search-engine libraries.

## Objectives

- Implement an indexing structure for fast retrieval.
- Implement phrase/substring and fuzzy search.
- Implement document similarity and ranking.
- Analyze time and space complexity and scalability choices.

# Hardware and Software Requirements

The abstract does not specify exact hardware, operating system, or programming language. The following are basic suggested requirements for implementing the described system.

## Hardware

- Laptop/Desktop computer
- Sufficient storage for the document corpus
- Keyboard and display for user interaction

## Software

- Programming environment/compiler
- Text/document corpus for indexing
- Development and testing environment for implementing algorithms from scratch

# Results

- The system provides a practical demonstration of information retrieval and search optimization.
- Exact keyword queries can be handled using an inverted index.
- Phrase/substring queries are processed using string-matching techniques.
- Fuzzy search uses edit distance to find closely matching terms.
- Document similarity uses term-frequency-based representations and similarity measures.
- The project analyzes the time and space complexity of the core algorithms.

Note: The supplied abstract does not contain numerical accuracy, timing, or benchmark values.

## Conclusion

- The Mini Search Engine demonstrates fundamental Data Structures and Algorithms through an information-retrieval application.
- Multiple query types are supported: exact, phrase/substring, fuzzy, and document similarity.
- From-scratch implementation helps expose the principles of indexing, matching, ranking, and similarity.
- Complexity analysis supports understanding of algorithmic efficiency and scalability.

## Future Scope

The abstract mentions scalability and appropriate algorithm choices, but it does not give a detailed future-work list. The following scope is derived directly from those stated themes and should be treated as proposed extensions.

- Improve scalability of indexing and retrieval for larger document collections.
- Compare alternative indexing and matching strategies.
- Evaluate time and space complexity on larger corpora.
- Improve ranking and similarity methods for more relevant results.
- Extend the system while preserving the from-scratch algorithmic focus.